Descriptive Statistics

The Importance of Visualization

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Slides can be accessed with the following link:

https://mluu921.github.io/cshs-fall-lecture-descriptive-statistics

Also available as a PDF on Onedrive

Why describe data?

- **Summarize** - reduce many observations into a few numbers or pictures a person can actually hold in their head
- **Detect anomalies** - outliers, data entry errors, and missingness patterns
- **Check assumptions** - most inferential methods (e.g. t-tests, linear regression) assume something about the shape of the data
- **Communicate** - Table 1 of nearly every clinical paper is descriptive statistics

Why do we need to visualize our data?

Data

dataset	x	y
A	55.3846	97.1795
A	51.5385	96.0256
A	46.1538	94.4872
A	42.8205	91.4103
A	40.7692	88.3333
A	38.7179	84.8718
A	35.641	79.8718
A	33.0769	77.5641
A	28.9744	74.4872
A	26.1538	71.4103

1–10 of 568 rows

Previous **1** 2 3 4 5 ... 57 Next

Let’s begin by taking descriptive measures

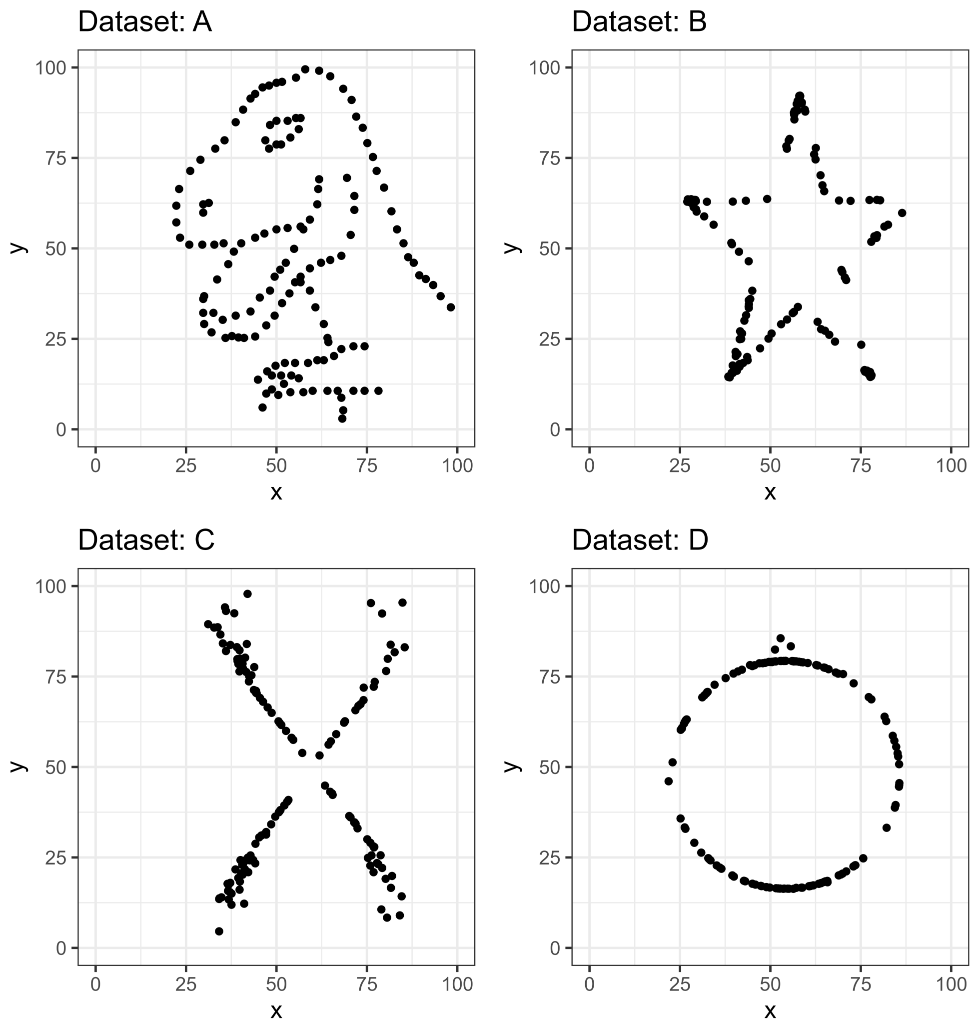
dataset	n	mean_x	sd_x	mean_y	sd_y
A	142	54.3	16.8	47.8	26.9
B	142	54.3	16.8	47.8	26.9
C	142	54.3	16.8	47.8	26.9
D	142	54.3	16.8	47.8	26.9

It appears the counts (n), mean (x), mean (y), and sd (x) and sd (y) are identical for ALL four datasets!

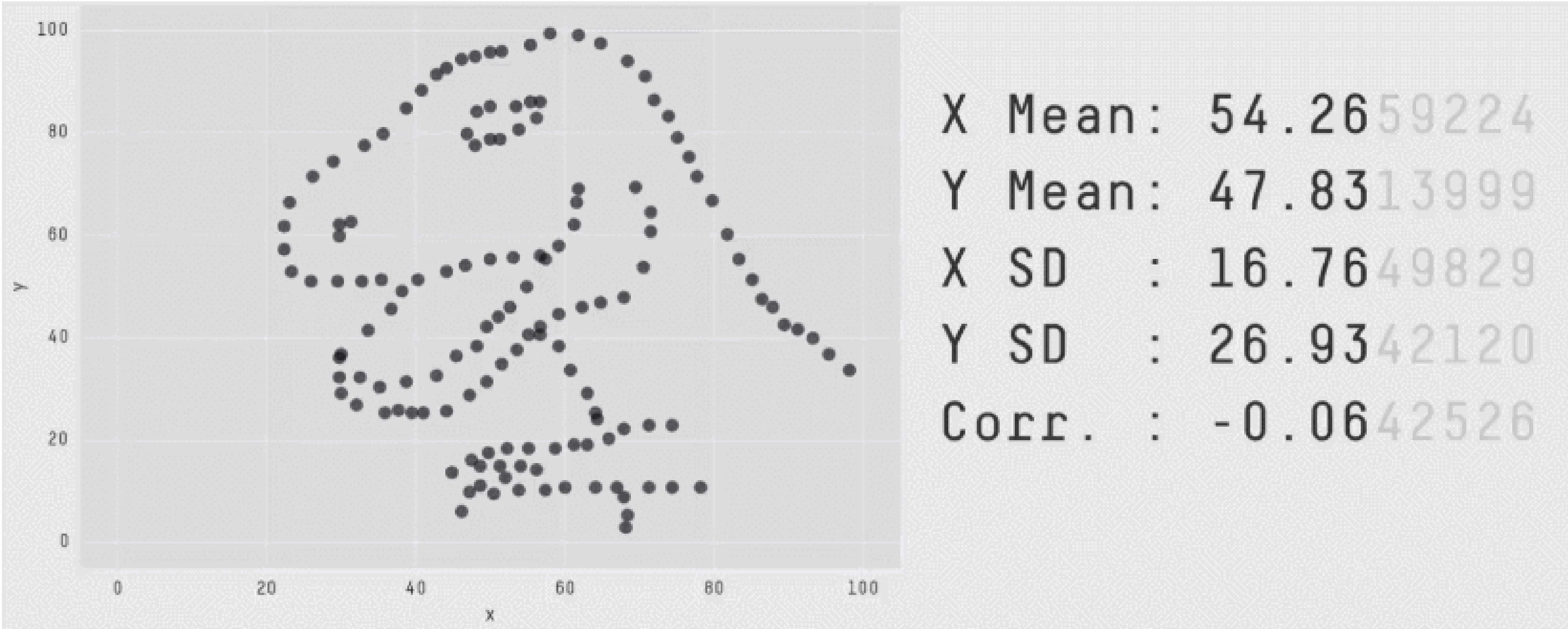
Can we conclude the datasets are similar or identical?

Not quite yet!

Let’s visualize the relationship of x and y



Dataset: A
Dataset: B
Dataset: C
Dataset: D
y
x
0
25
50
75
100


X Mean: 54.2659224
Y Mean: 47.8313999
X SD : 16.7649829
Y SD : 26.9342120
Corr. : -0.0642526
x
y
0
20
40
60
80
100

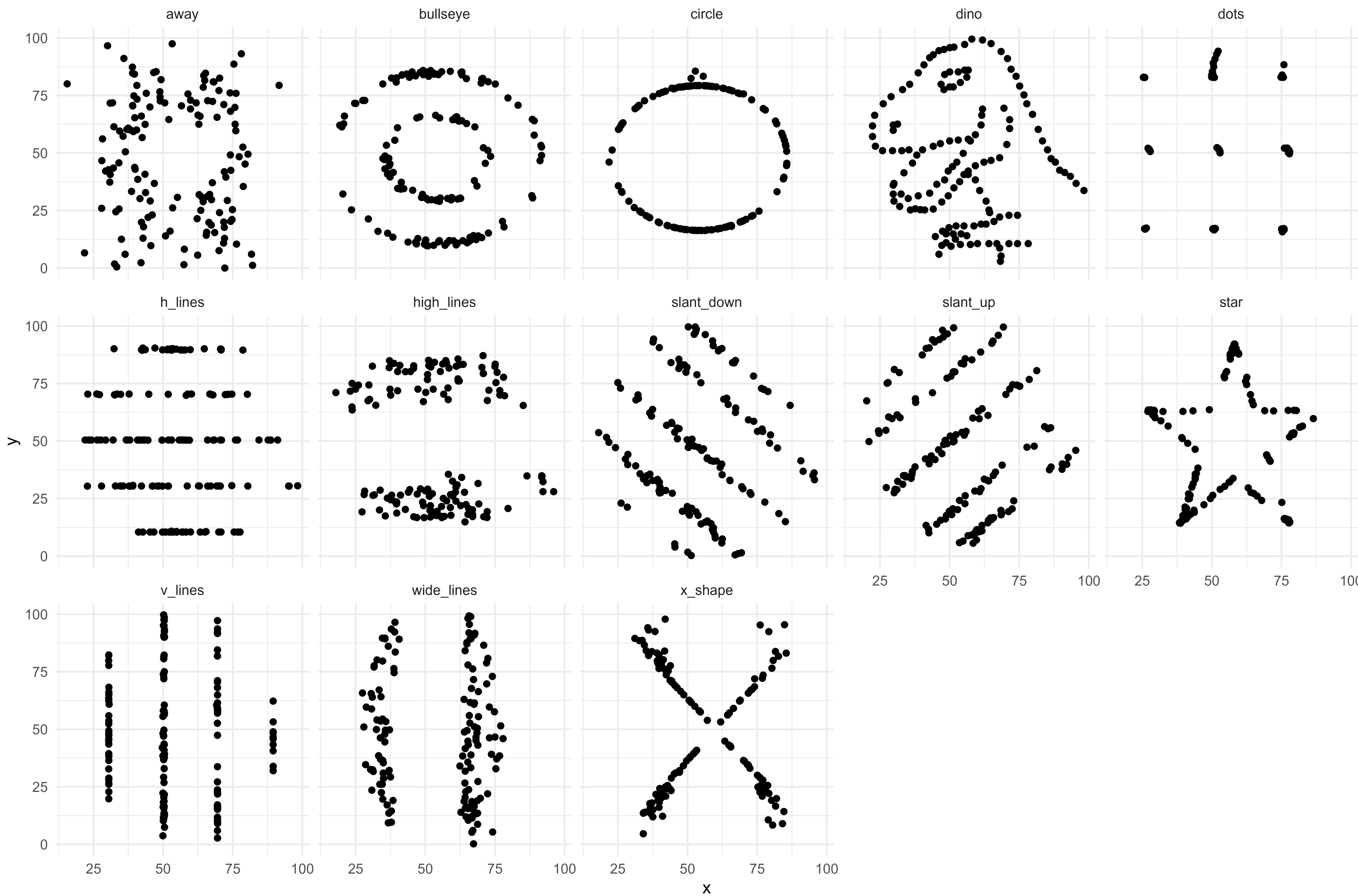
Although simple quantitative summaries are similar ...

They can appear drastically different when visualized!

Datasaurus Dozen

- The original “Datasaurus” or “dino” was created by **Alberto Cairo** in the following blog post[1]
- He was then later popularized by the paper published by **Justin Matejka** and **George Fitzmaurice**, titled ‘Same Stats, Different Graphs: Generating Datasets with Varied Appearance and Identical Statistics through Simulated Annealing’[2], where they simulated 12 additional datasets in addition to the original “Datasaurus” with nearly identical simple statistics

Datasaurus Dozen

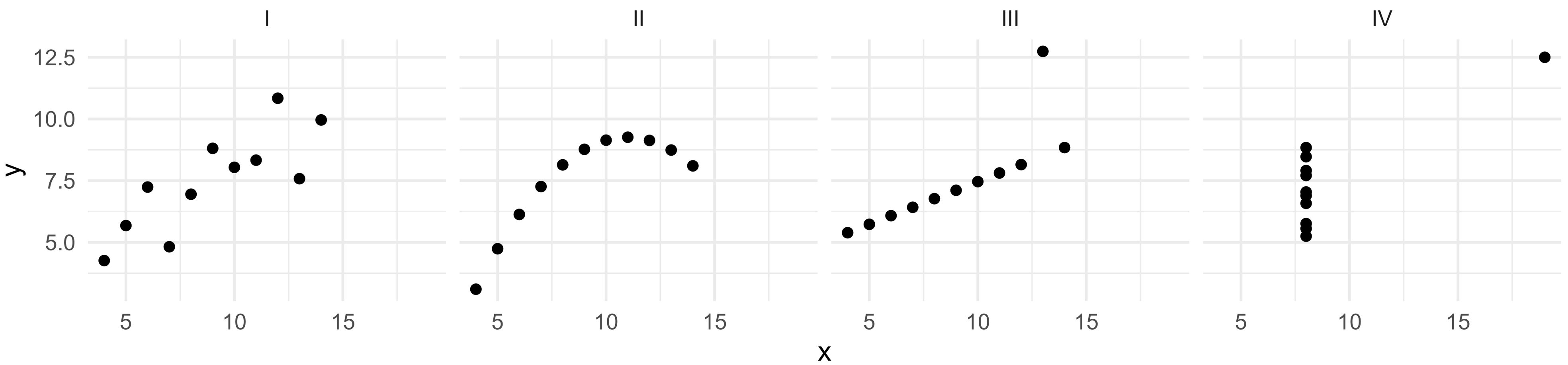


Anscombe Quartet

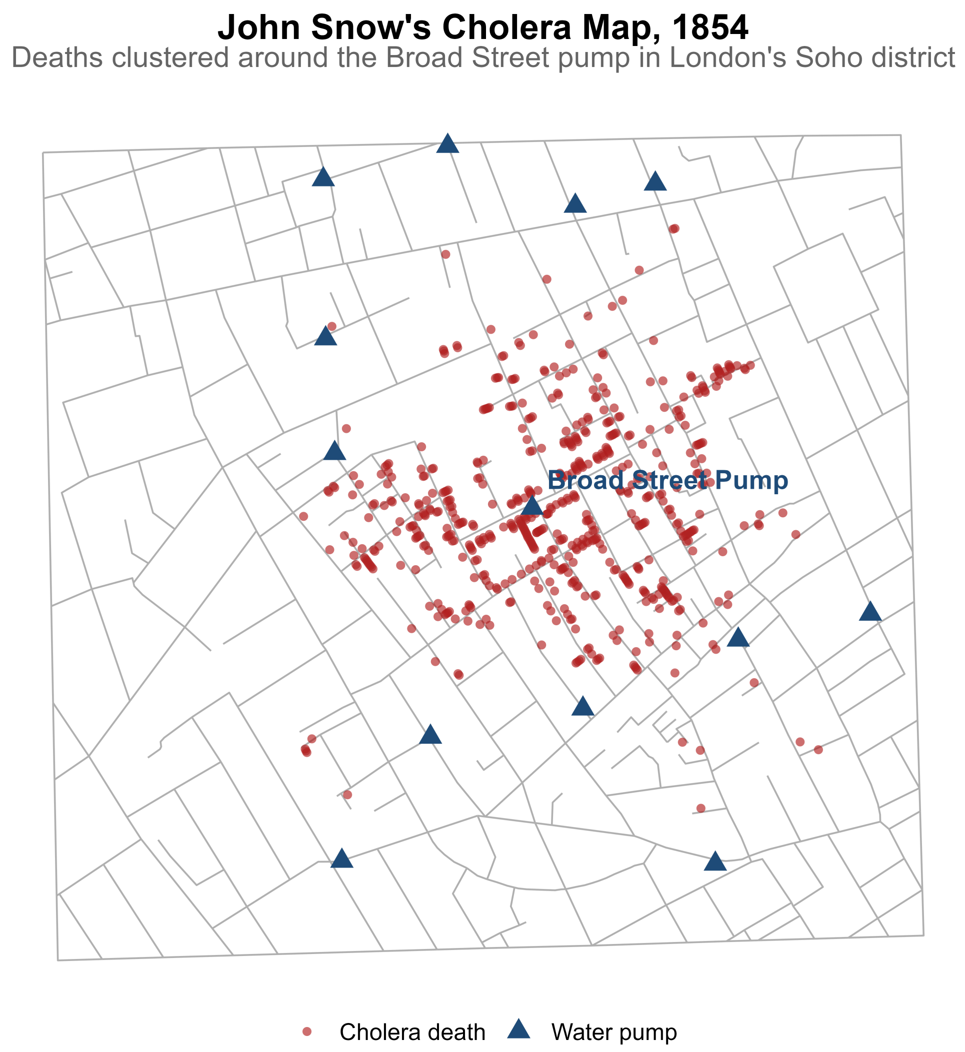
- The datasaurus dozen is a modern take on the classical "Anscombe's Quartet"[1]
- Comprised of four datasets that have nearly identical simple summary measures, yet have very different distributions and appear vastly different when plotted

Anscombe Quartet

dataset	n	mean_x	sd_x	mean_y	sd_y
I	11.00	9.00	3.32	7.50	2.03
II	11.00	9.00	3.32	7.50	2.03
III	11.00	9.00	3.32	7.50	2.03
IV	11.00	9.00	3.32	7.50	2.03



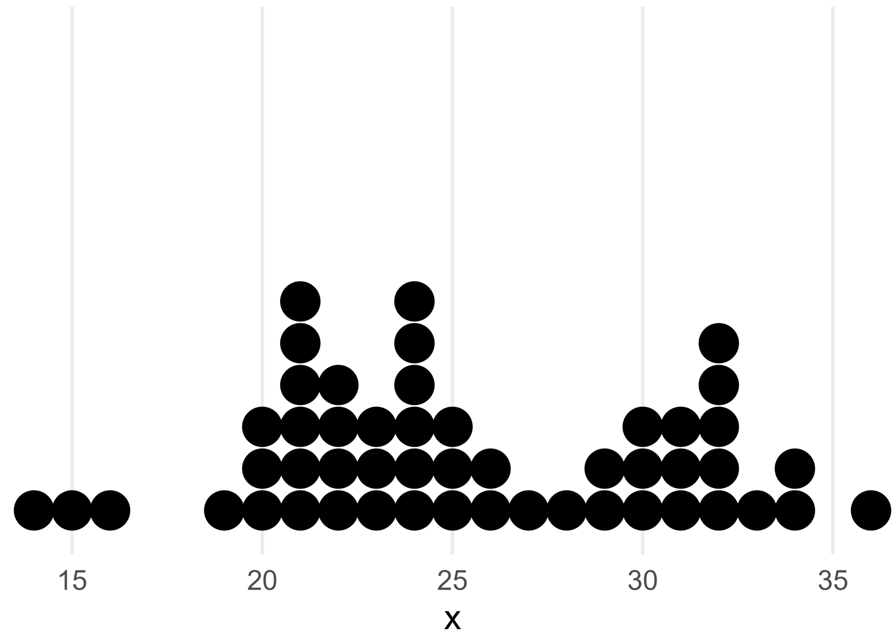
A Historical Case for Visualization



Types of Graphical Visualizations

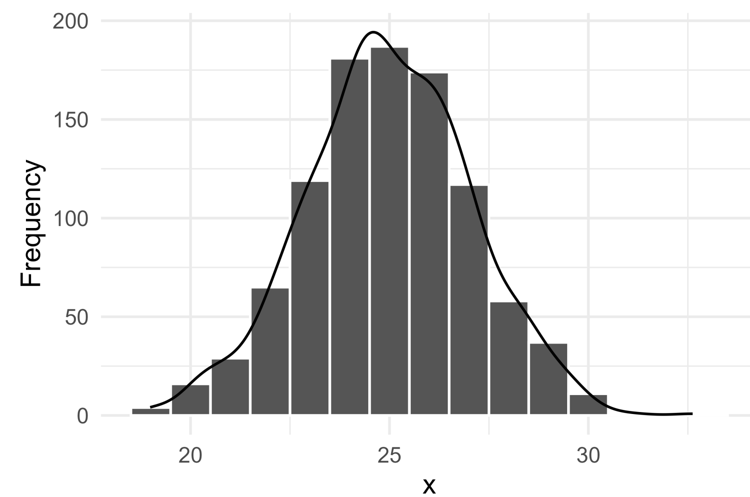
Dot plot

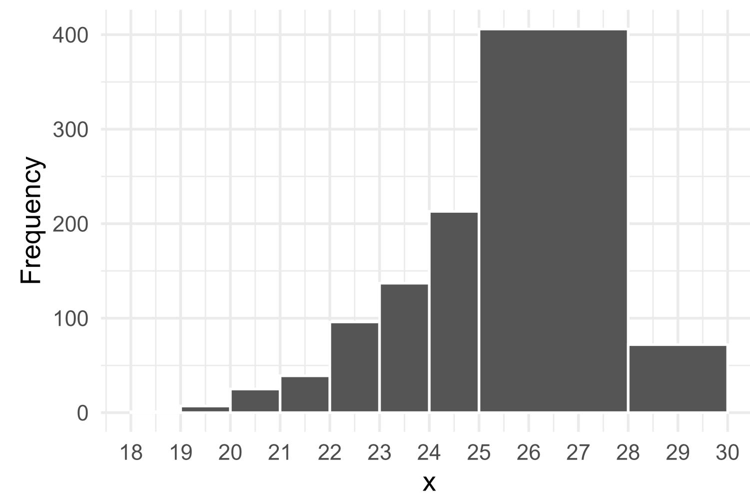
- Useful for small to moderate sized data
- Allows us to visualize the spread and distribution of a single continuous or discrete variable
  - e.g. length of stay
- The X axis is the variable of interest and each dot represents a single observation
- Easy to identify the mode
- Highlights clusters, gaps, and outliers
- Intuitive and easy to understand



Histogram

- Useful for all sized data (small and large)
- Allows us to visualize the spread and distribution of continuous variables
- Each bar represents a ‘bin’ or a defined interval of values
- Although not as common, the width of the bins does NOT have to be equal!
- The y axis or the height of the bar represents the count of the number of values that fall into each bin
- The y axis is also commonly normalized to ‘relative’ frequencies to show the proportion of cases or density that falls into each bin.

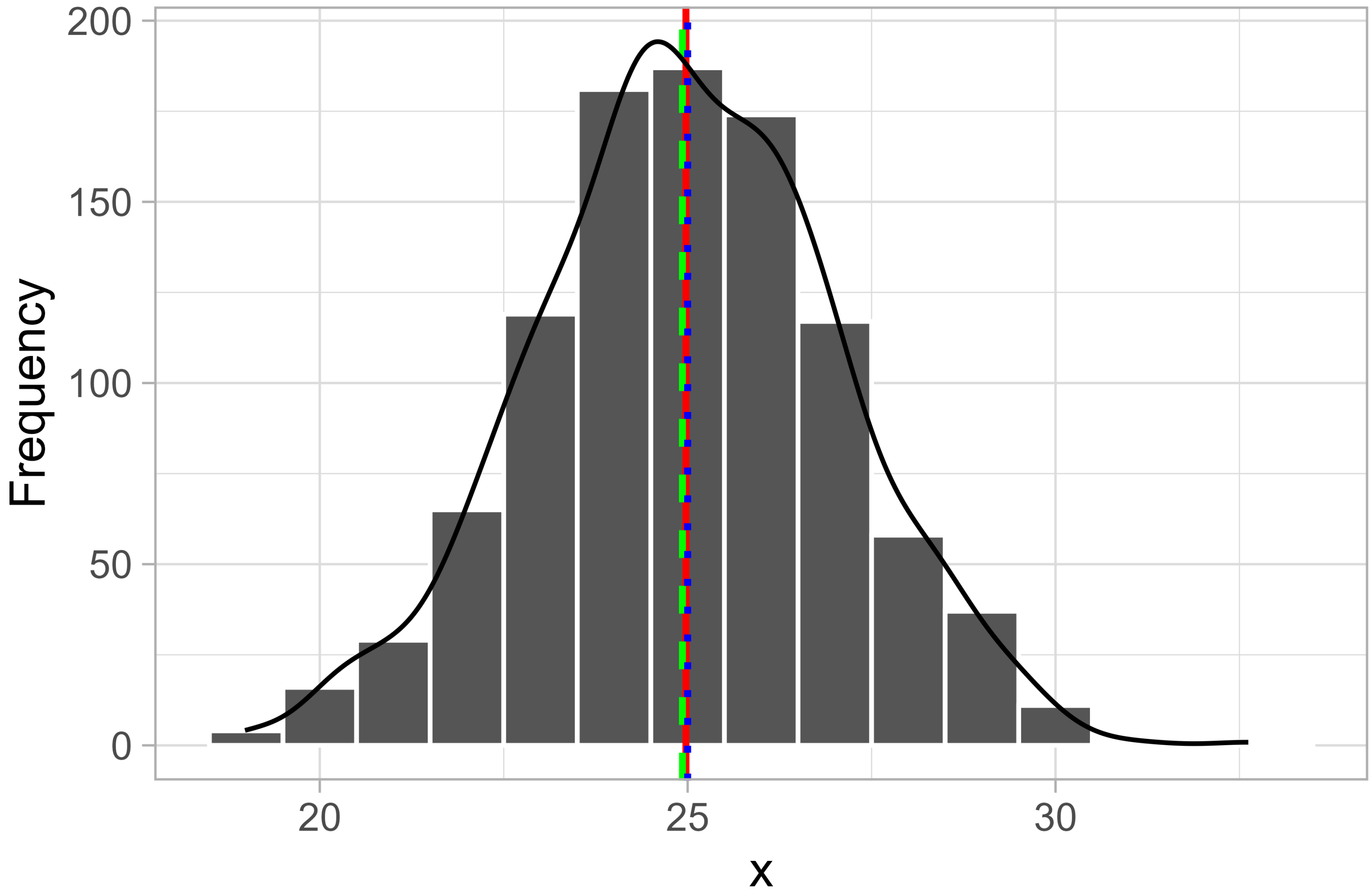




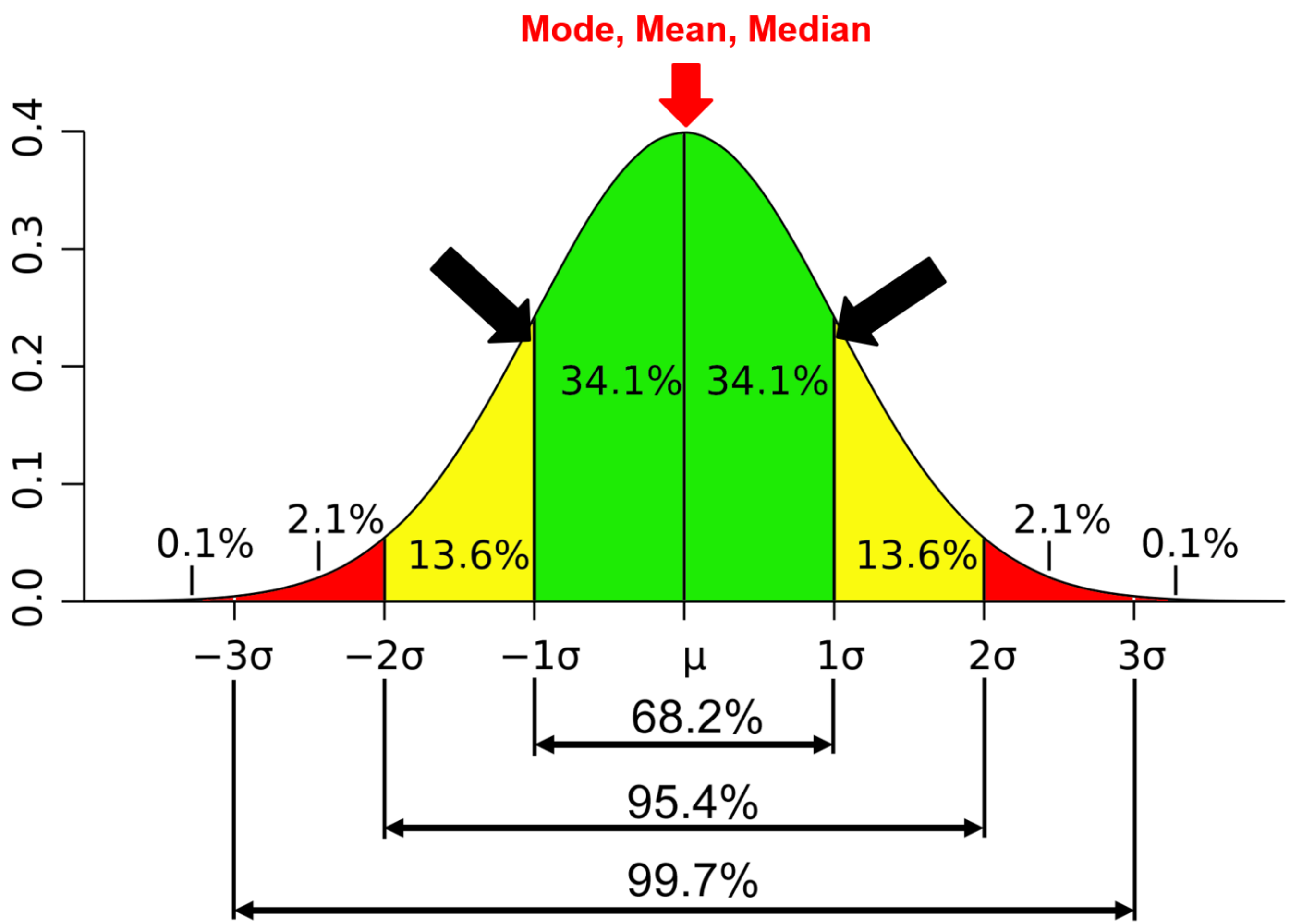
Distribution

> "A distribution is simply a collection of data, or scores, on a variable. Usually, these scores are arranged in order from smallest to largest and then they can be presented graphically."[1]

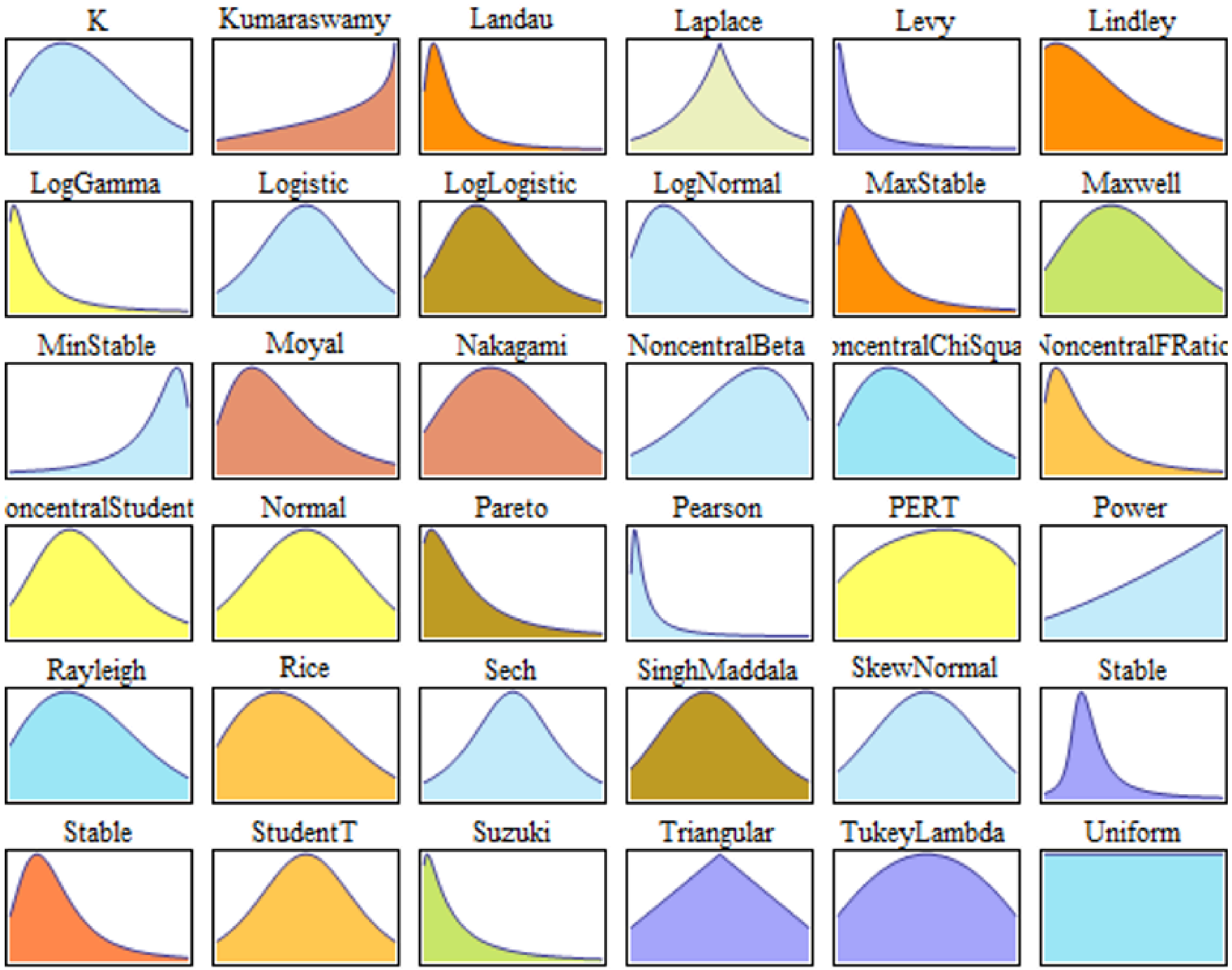
Distribution



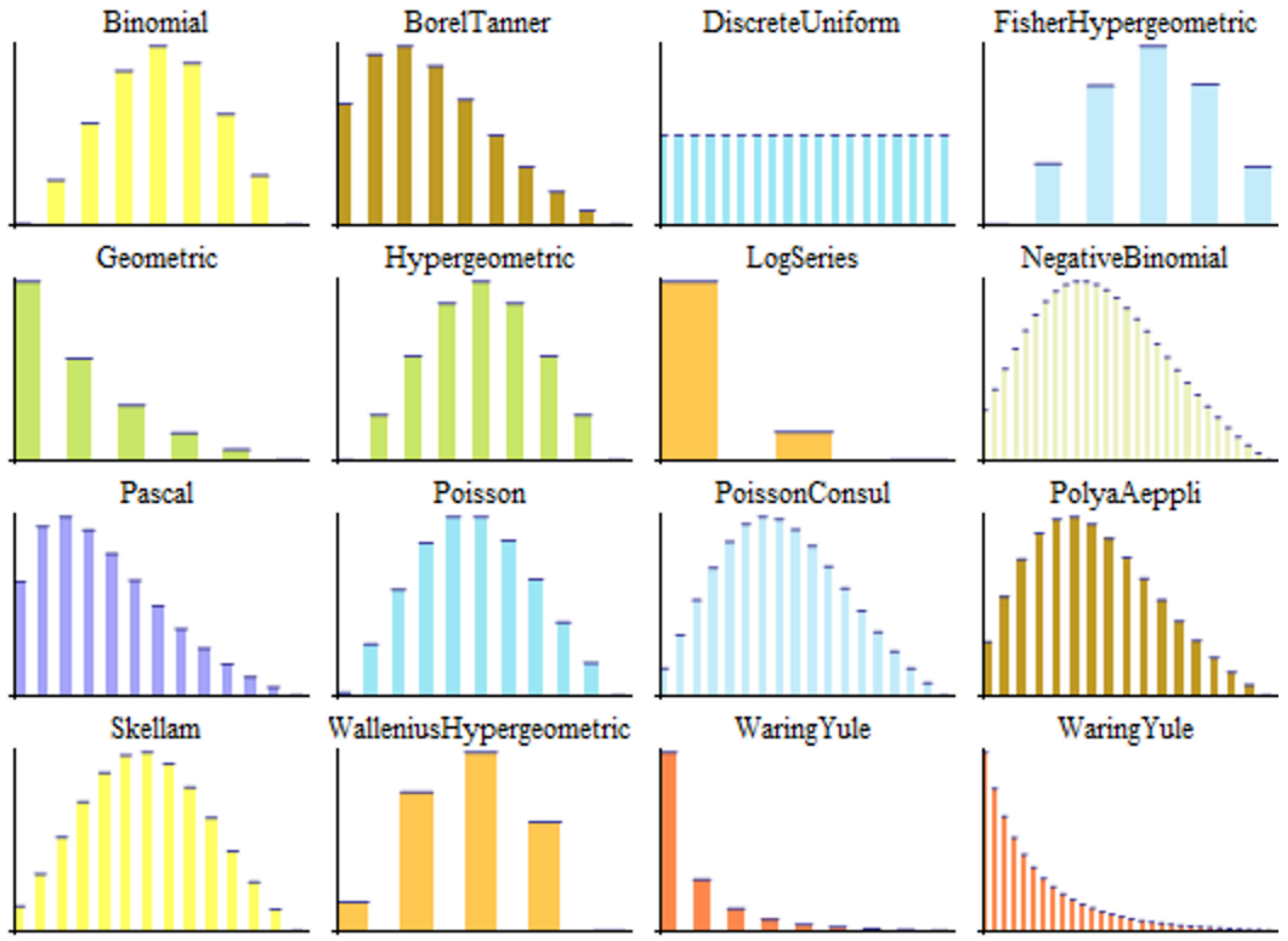
Normal Distribution



Univariate Continuous Distributions

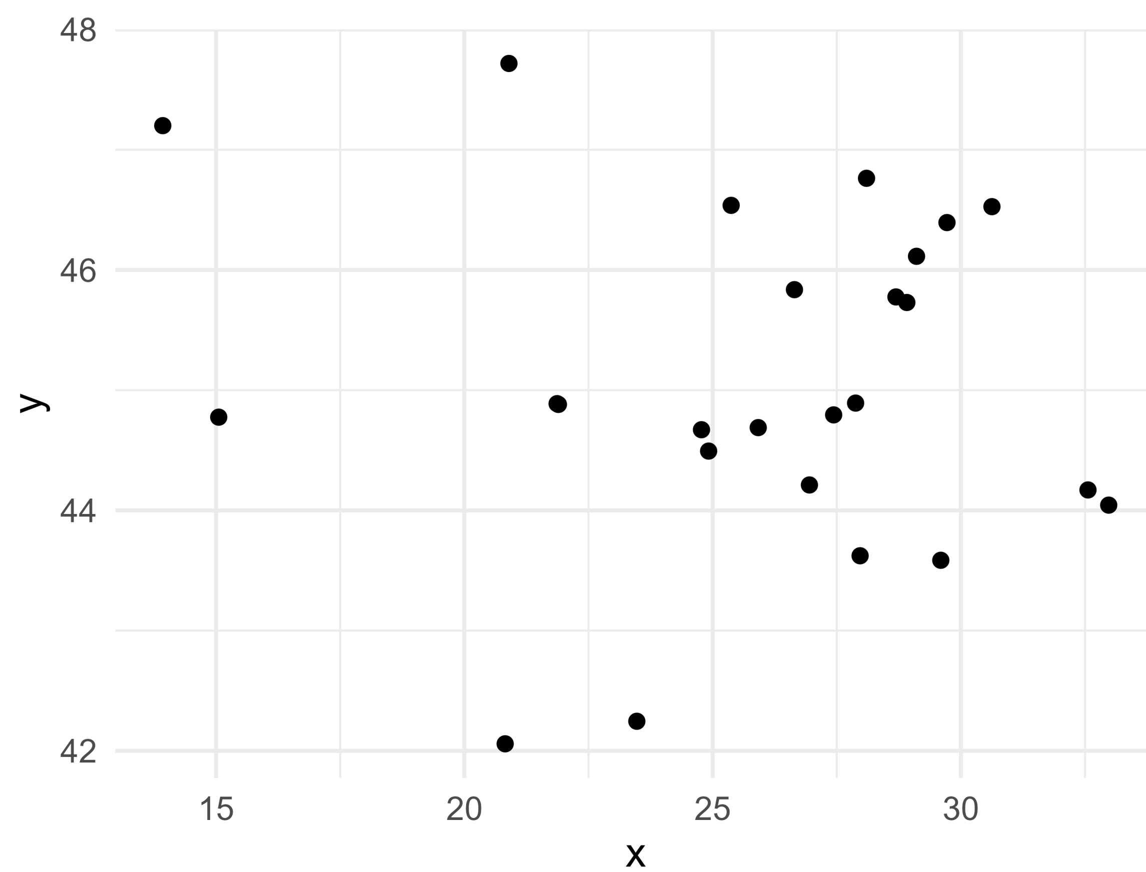


Univariate Discrete Distributions



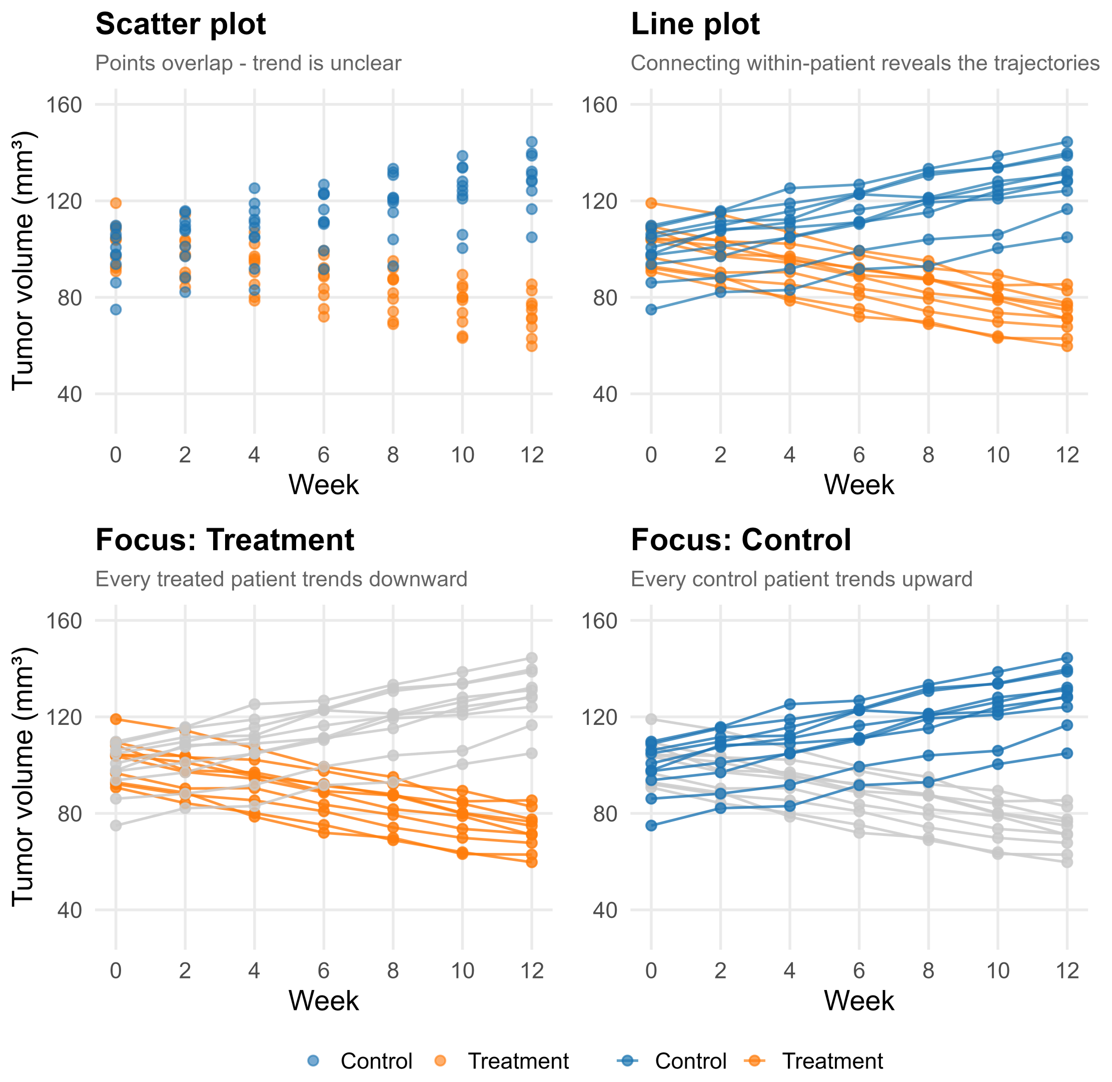
Scatter plot

- Used to visualize the relationship between two continuous variables
- Useful for detecting patterns that are obscured from quantitative summaries like what we observed in Anscombe's quartet and the Datasaurus dozen.



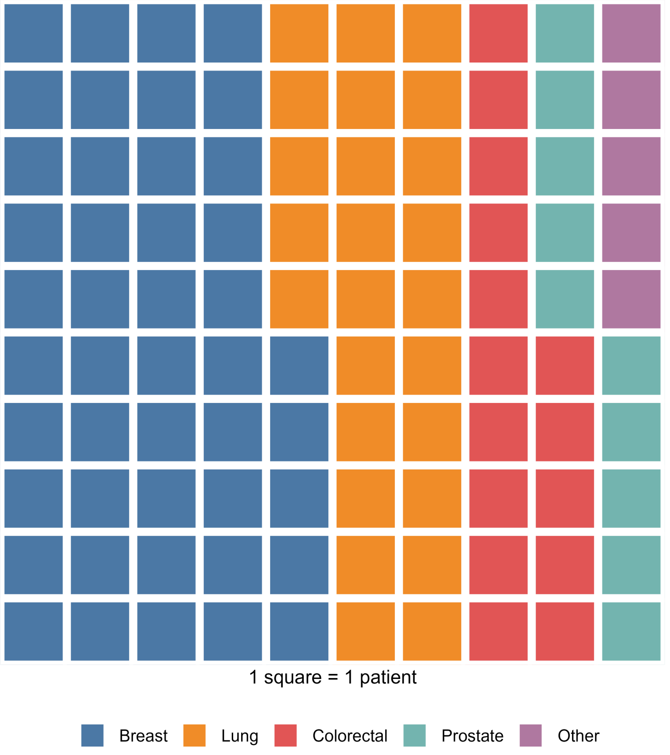
Line plot

- Used to visualize a **continuous** variable over an ordered sequence - typically time
- Points are connected in order, revealing trends and patterns of change that a scatter plot alone would obscure
- Common in clinical research for tracking biomarkers, tumor burden, or patient-reported outcomes over follow-up
- Multiple lines allow direct comparison between groups (e.g. treatment arms)



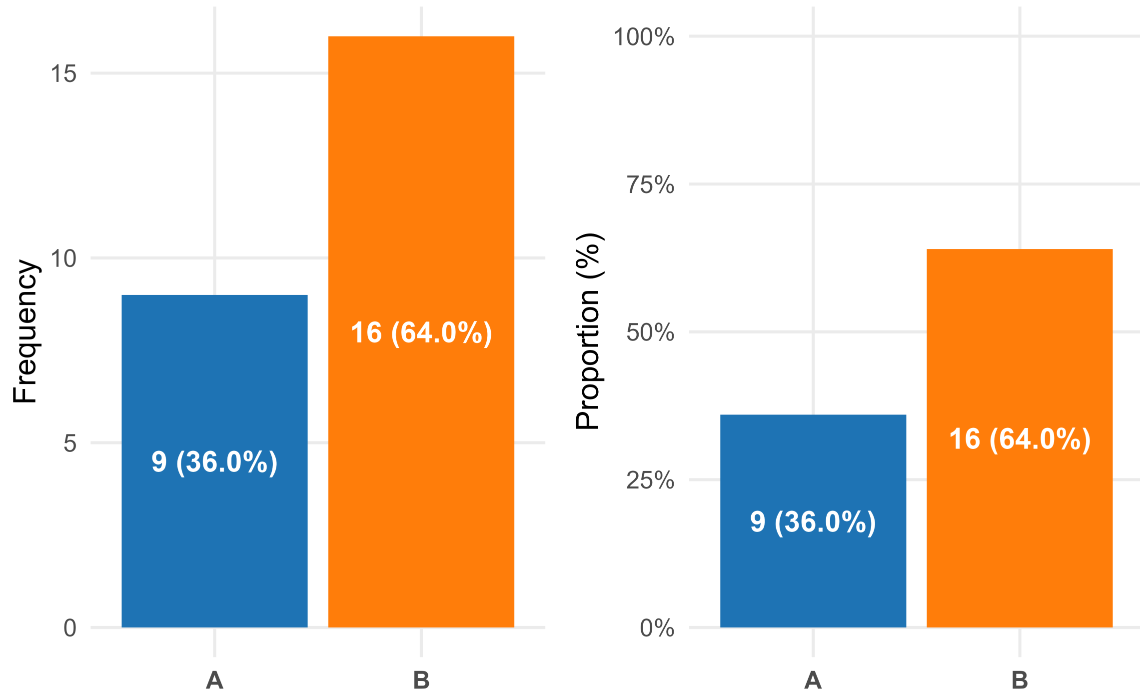
Waffle plot

- Useful for visualizing **categorical** data, suitable for small to moderate sized data.
- Best used when you want to emphasize the "part-to-whole" relationship in your data
- Each square represents a fixed number of observations
- Effective for communicating data to non-technical audiences



Bar plot

- Useful for visualizing categorical data
- Commonly used to present counts and proportion of each level
- Allows us to quickly observe the difference in magnitude of each level based on the height of each bar

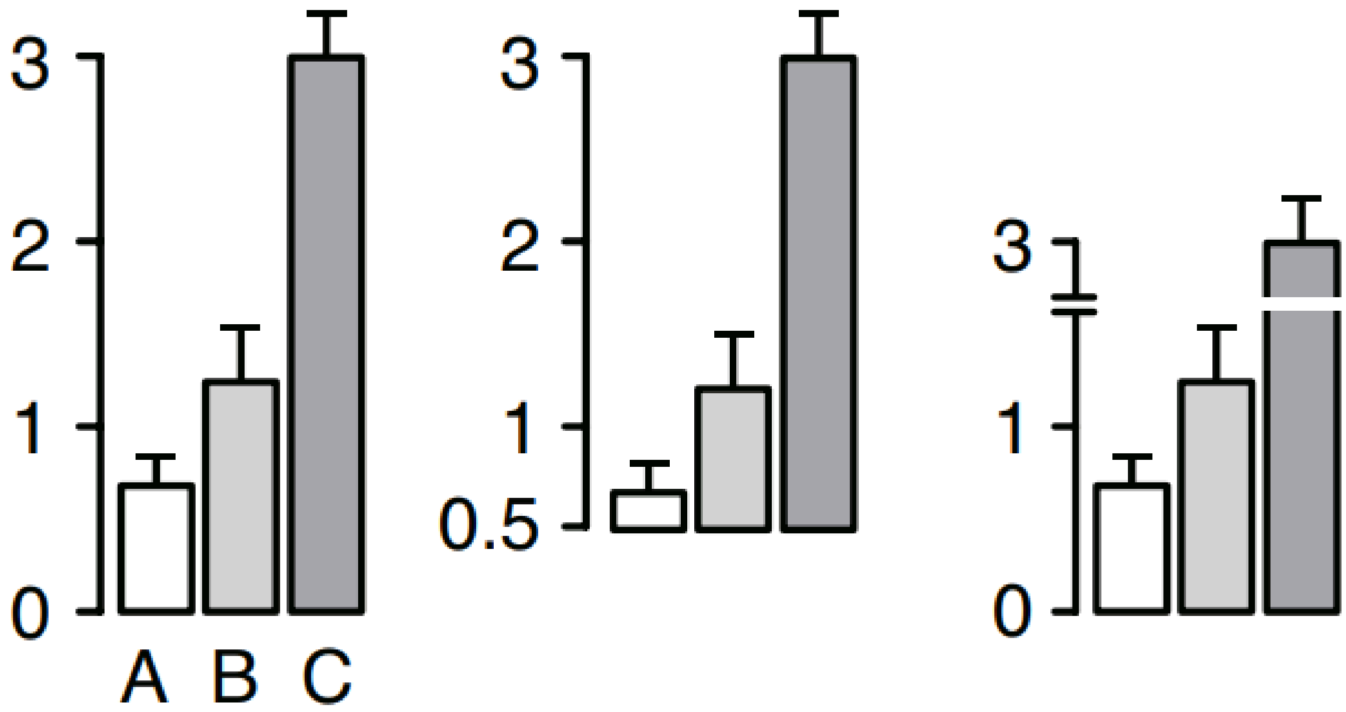


However…

Bar plots are commonly misused!

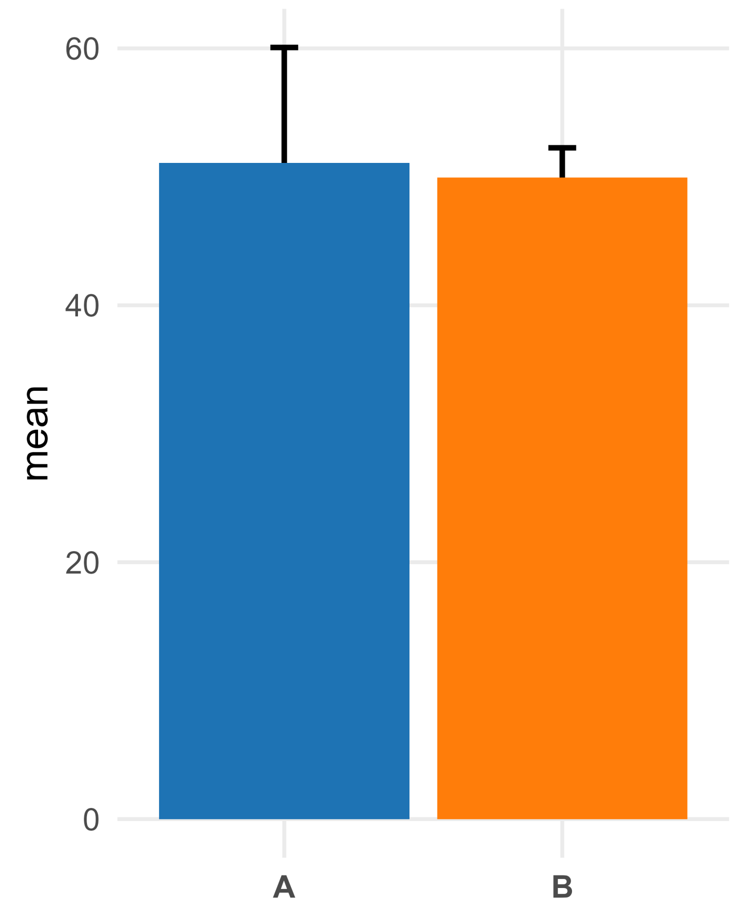
How NOT to Bar Plot

Not recommended



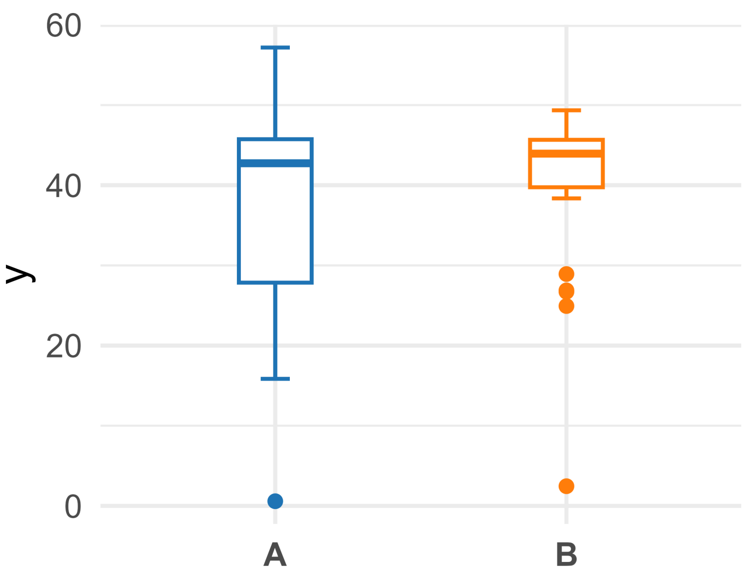
How NOT to Bar Plot: Dynamite Plots

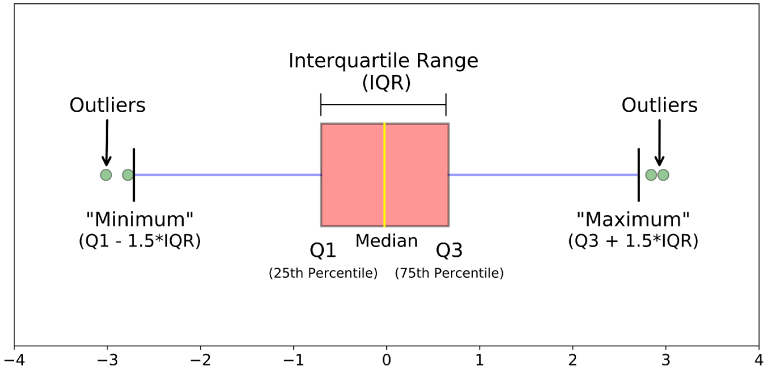
- Although frequently found and prevalent in the literature, this is NOT to be used to describe mean and dispersion (continuous data)
- Only shows one arm of the error bar, making overlap comparisons difficult
- Promotes misconception of the mean being related to its height rather the position of the top of the bar
- Obscures the distribution and spread of the data



Box plot

- Useful for describing **continuous** variables following a uni-modal distribution
  - e.g. a single peak
- The box is representative of common quantitative measures
  - Top of box is the 75th quantile
  - Middle dash inside box is the 50th quantile
  - Bottom of box is the 25th quantile
  - Width of the box is the interquartile range (IQR)
- The 'whiskers' are artificial 'fences' that helps identify potential outliers in the data
  - Defined as Q1 - 1.5*IQR and Q3 + 1.5*IQR

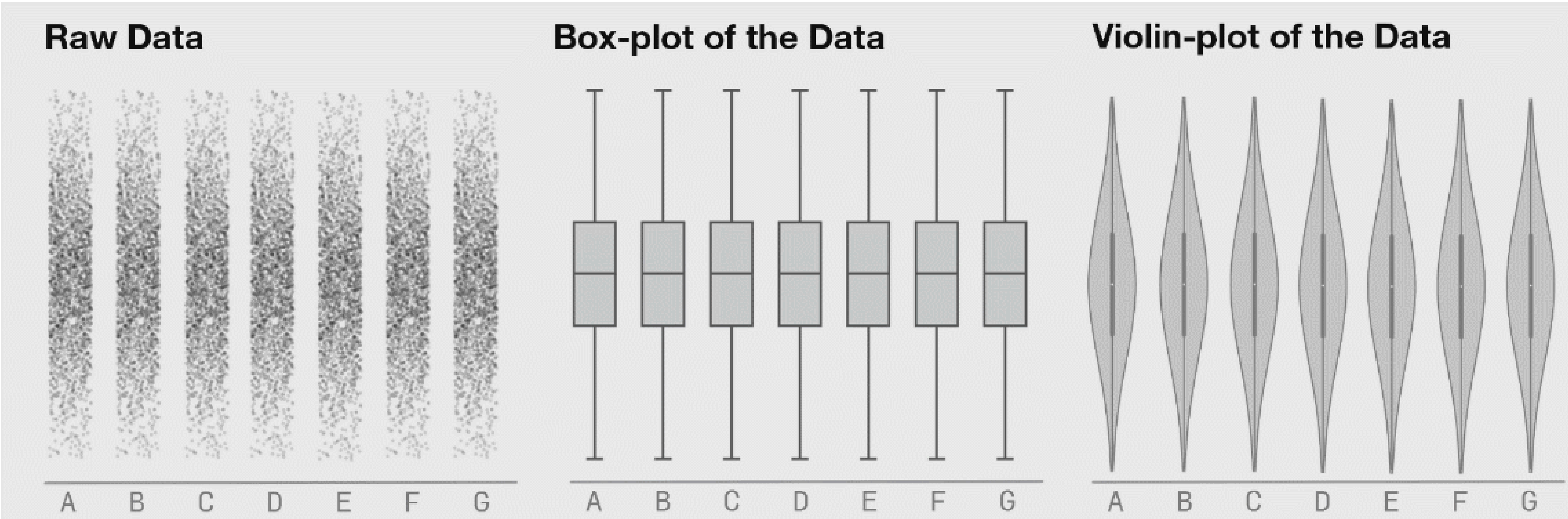




What are some of the problems with a box plot?

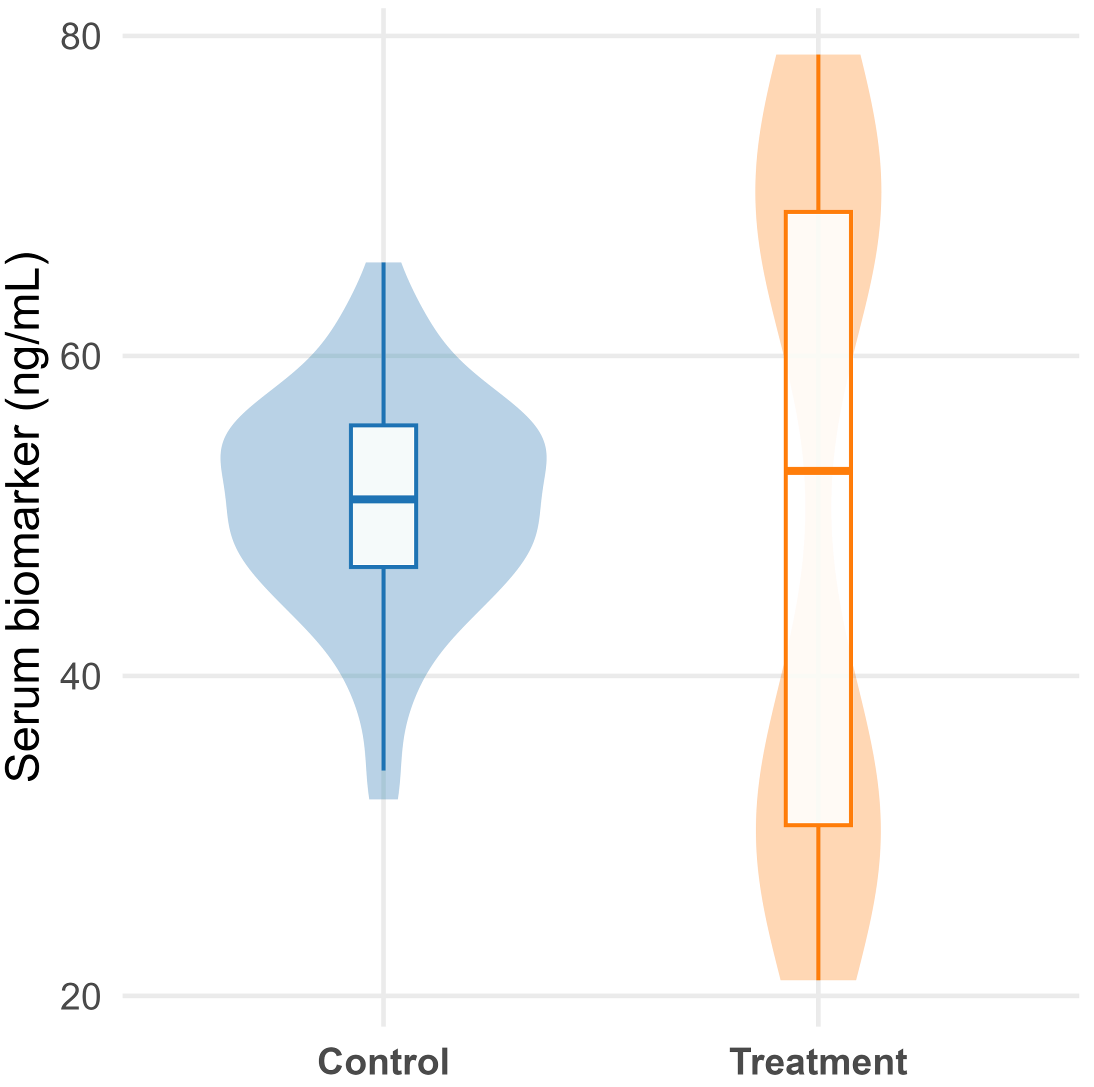
They are based on quantitative summaries!

Box plot


Raw Data
A B C D E F G
Box-plot of the Data
A B C D E F G
Violin-plot of the Data
A B C D E F G

Violin plot

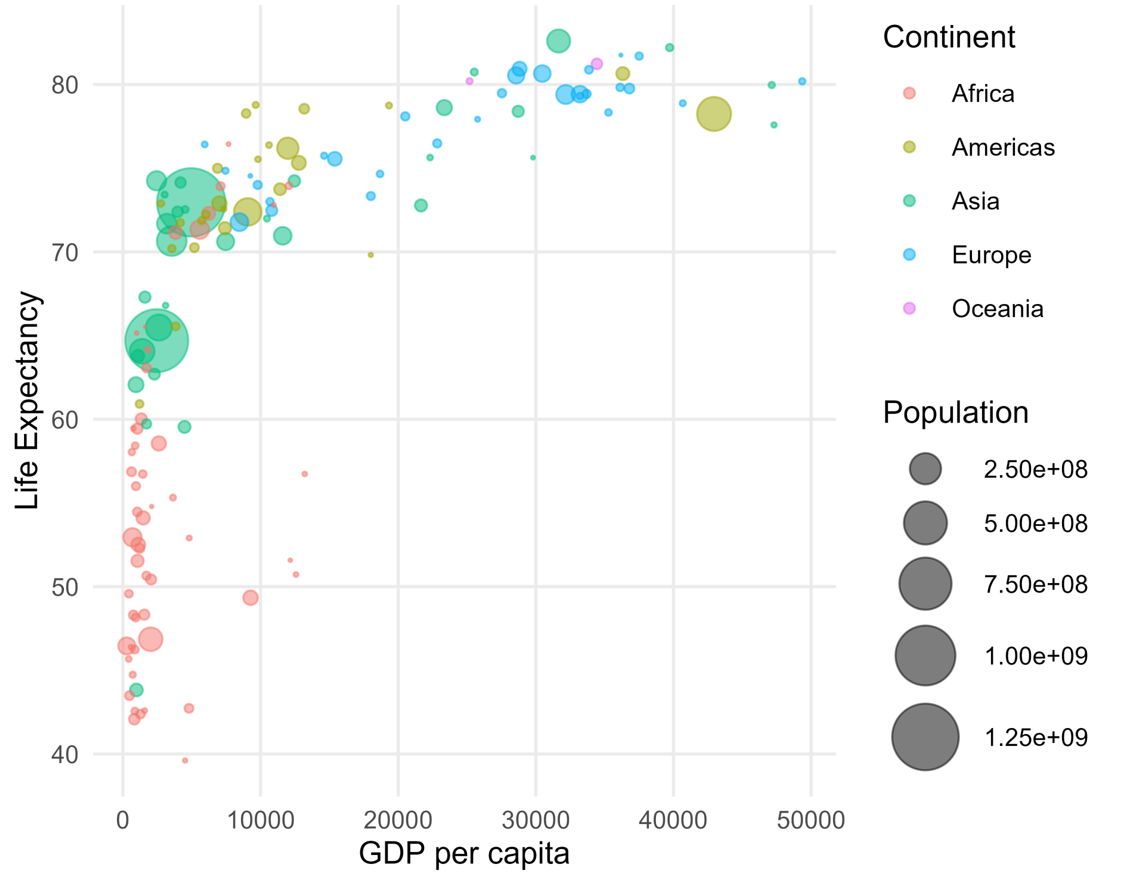
- Violin plots are box plots, with an overlay of the density distribution (histogram) of the data
- More informative than a simple box plot
- Visualizes the full distribution of the data
- Especially useful for bimodal or multimodal distribution
  - e.g. distribution of data with multiple peaks



Beyond two variables

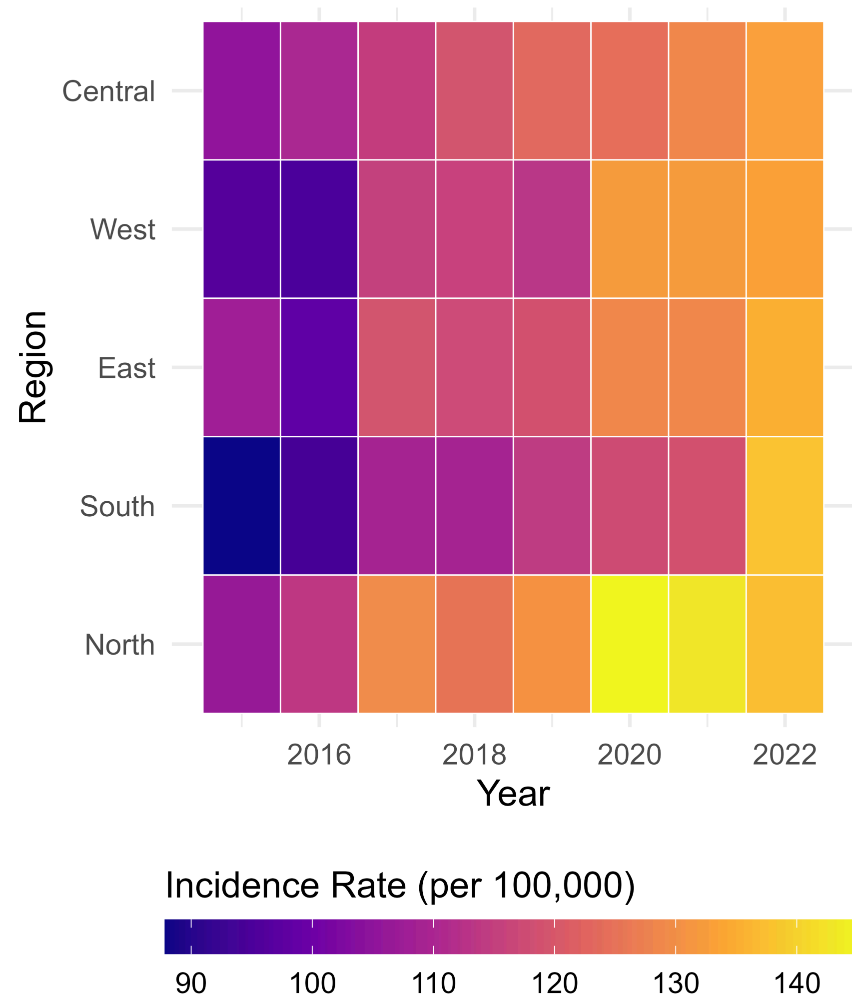
Bubble Plot

- Visualizes the relationship between two **continuous** variables, with a third **continuous** variable represented by the size of the bubbles.
- The color of the bubbles can represent a fourth **categorical** or **continuous** variable.
- Useful for identifying patterns, trends, and clusters in multivariate data.



Tile Plots / Heatmaps

- Used to visualize the relationship between two **categorical** variables, with a **continuous** variable represented by color intensity.
- The color gradient can represent a summary statistic (e.g., mean, median, count) for each combination of the two categorical variables.
- Useful for identifying patterns, trends, and outliers in multivariate categorical data.



Summary

- **One continuous variable**
  - Dot plot
  - Histogram
  - Box plot
  - Violin plot
- **One or more categorical variable**
  - Bar plot
  - Waffle plot
- **Two continuous variable**
  - Scatter plot
- **One continuous by categorical variable**
  - Dot plot
  - Box plot
  - Violin plot
- **Continuous variable over time**
  - Line plot
- **Beyond two variables**
  - Bubble Plot
  - Tile Plot / Heatmap

Descriptive summaries are useful, however …

Don’t forget to visualize your data!

Questions